

1. Create a bash script to **check if a directory is available or not**.

- ➔ Created check_dir.bash.
- ➔ Used the -d condition to check the directory.
- ➔ Added an if-else statement to display the result.
- ➔ Made the script executable using chmod +x.
- ➔ Executed the script and verified the output.

```
root@ip-172-31-25-75:~/poojith/scripts
[root@ip-172-31-25-75 scripts]# vi check_dir.bash
[root@ip-172-31-25-75 scripts]# chmod 755 check_dir.bash
[root@ip-172-31-25-75 scripts]# cat check_dir.bash
#!/bin/bash

ls -l /root/poojith/scripts/sample

if [ $? -eq 0 ]
then
    echo "Directory Exists"
else
    echo "Directory does not exist"
fi

[root@ip-172-31-25-75 scripts]# ./check_dir.bash
total 0
Directory Exists
[root@ip-172-31-25-75 scripts]# ls
a      c      current_day.bash  error_exist.bash  hello_world.bash  perform_task.bash  run_numberof_times.bash  sequence.bash
admin.bash  case_scripts.bash  define_tasks.bash  error_file.bash  list_of_users.bash  read_input.bash  sample      specify_days.bash
b      check_dir.bash  define_var.bash  exit_status1.bash  loop_output1.bash  response_output.bash  sample_input.bash  variable.bash
basic_second.bash  create_files.bash  del_files.bash  exit_status2.bash  loop_output2.bash  run_commands_within.bash  sample_input2.bash  variable_value_met.bash
[root@ip-172-31-25-75 scripts]# |
```

2. Create a bash script to **create multiple files**.

- ➔ Created a file multi_files.bash
- ➔ Used touch command to create multiple files
- ➔ Changed permissions to the file using chmod
- ➔ Executed and verified the output.

```
root@ip-172-31-25-75:~/poojith/scripts
[root@ip-172-31-25-75 scripts]# vi multi_files.bash
[root@ip-172-31-25-75 scripts]# cat multi_files.bash
#!/bin/bash

touch { A..F }

[root@ip-172-31-25-75 scripts]# ./multi_files.bash
[root@ip-172-31-25-75 scripts]# ls
A..F  basic_second.bash  create_files.bash  error.txt  exit_status3.bash  hello_world.bash  loop_output1.bash  loop_output2.bash  perform_task.bash  read_input.bash  response_output.bash  sample      sample_input.bash  sample_input2.bash  sequence.bash  specify_days.bash  variable.bash  variable_value_met.bash
admin.bash  case_scripts.bash  current_day.bash  error_exist.bash  error_file.bash  list_of_users.bash  multi_files.bash  mon_tue.bash  run_commands_within.bash  run_numberof_seconds.bash  run_numberof_times.bash  run_numberof_times.bash  sample      sample_input.bash  sample_input2.bash  sequence.bash  specify_days.bash  variable.bash  variable_value_met.bash
b      check_dir.bash  define_tasks.bash  define_var.bash  exit_status1.bash  exit_status2.bash  error.txt  error_exist.bash  error_file.bash  hello_world.bash  list_of_users.bash  loop_output1.bash  loop_output2.bash  perform_task.bash  read_input.bash  response_output.bash  sample      sample_input.bash  sample_input2.bash  sequence.bash  specify_days.bash  variable.bash  variable_value_met.bash
[root@ip-172-31-25-75 scripts]# ./multi_files.bash
[root@ip-172-31-25-75 scripts]# ls
1  6  D  b  check_variable.bash  create_files.bash  del_files.bash  error.txt  error_exist.bash  exit_status1.bash  exit_status2.bash  exit_status3.bash  hello_world.bash  loop_output1.bash  loop_output2.bash  perform_task.bash  read_input.bash  response_output.bash  sample      sample_input.bash  sample_input2.bash  sequence.bash  specify_days.bash  variable.bash  variable_value_met.bash
2  A..F  c  basic_second.bash  current_day.bash  define_tasks.bash  define_var.bash  error_file.bash  error_exist.bash  error_file.bash  hello_world.bash  list_of_users.bash  loop_output1.bash  loop_output2.bash  perform_task.bash  read_input.bash  response_output.bash  sample      sample_input.bash  sample_input2.bash  sequence.bash  specify_days.bash  variable.bash  variable_value_met.bash
4  D  c  case_scripts.bash  create_files.bash  del_files.bash  error.txt  error_exist.bash  error_file.bash  hello_world.bash  list_of_users.bash  loop_output1.bash  loop_output2.bash  perform_task.bash  read_input.bash  response_output.bash  sample      sample_input.bash  sample_input2.bash  sequence.bash  specify_days.bash  variable.bash  variable_value_met.bash
5  admin.bash  check_dir.bash  define_tasks.bash  define_var.bash  exit_status1.bash  exit_status2.bash  error.txt  error_exist.bash  error_file.bash  hello_world.bash  list_of_users.bash  loop_output1.bash  loop_output2.bash  perform_task.bash  read_input.bash  response_output.bash  sample      sample_input.bash  sample_input2.bash  sequence.bash  specify_days.bash  variable.bash  variable_value_met.bash
[root@ip-172-31-25-75 scripts]# ./multi_files.bash
[root@ip-172-31-25-75 scripts]# ls
1  6  D  b  check_variable.bash  create_files.bash  del_files.bash  error.txt  error_exist.bash  exit_status1.bash  exit_status2.bash  exit_status3.bash  hello_world.bash  loop_output1.bash  loop_output2.bash  perform_task.bash  read_input.bash  response_output.bash  sample      sample_input.bash  sample_input2.bash  sequence.bash  specify_days.bash  variable.bash  variable_value_met.bash
2  A..F  c  basic_second.bash  current_day.bash  define_tasks.bash  define_var.bash  error_file.bash  error_exist.bash  error_file.bash  hello_world.bash  list_of_users.bash  loop_output1.bash  loop_output2.bash  perform_task.bash  read_input.bash  response_output.bash  sample      sample_input.bash  sample_input2.bash  sequence.bash  specify_days.bash  variable.bash  variable_value_met.bash
4  D  c  case_scripts.bash  create_files.bash  del_files.bash  error.txt  error_exist.bash  error_file.bash  hello_world.bash  list_of_users.bash  loop_output1.bash  loop_output2.bash  perform_task.bash  read_input.bash  response_output.bash  sample      sample_input.bash  sample_input2.bash  sequence.bash  specify_days.bash  variable.bash  variable_value_met.bash
5  admin.bash  check_dir.bash  define_tasks.bash  define_var.bash  exit_status1.bash  exit_status2.bash  error.txt  error_exist.bash  error_file.bash  hello_world.bash  list_of_users.bash  loop_output1.bash  loop_output2.bash  perform_task.bash  read_input.bash  response_output.bash  sample      sample_input.bash  sample_input2.bash  sequence.bash  specify_days.bash  variable.bash  variable_value_met.bash
[root@ip-172-31-25-75 scripts]# cat multi_files.bash
#!/bin/bash

touch { A..E }
```

3. Create a bash script to **take a backup of a directory**.

- ➔ Created a file
- ➔ Used tar command to compress a directory and store it as a backup file.
- ➔ Changed permissions of file.
- ➔ Executed and verified the output.

```
root@ip-172-31-25-75:~/poojith/scripts
[root@ip-172-31-25-75 scripts]# vi backup.bash
[root@ip-172-31-25-75 scripts]# chmod 755 backup.bash
[root@ip-172-31-25-75 scripts]# cat backup.bash
#!/bin/bash

tar cvf sample.tar sample

ls

[root@ip-172-31-25-75 scripts]# ./backup.bash
sample/
1  6  D  b  check_dir.bash  define_var.bash  exit_status1.bash  loop_output1.bash  read_input.bash  sample      sequence.bash  specify_days.bash  variable.bash  variable_value_met.bash
2  A..F  c  basic_second.bash  current_day.bash  error_exist.bash  error_file.bash  hello_world.bash  list_of_users.bash  multi_files.bash  perform_task.bash  response_output.bash  sample      sample_input.bash  sample_input2.bash  sequence.bash  specify_days.bash  variable.bash  variable_value_met.bash
4  D  c  case_scripts.bash  create_files.bash  del_files.bash  error.txt  error_exist.bash  error_file.bash  hello_world.bash  list_of_users.bash  loop_output1.bash  loop_output2.bash  perform_task.bash  read_input.bash  response_output.bash  sample      sample_input.bash  sample_input2.bash  sequence.bash  specify_days.bash  variable.bash  variable_value_met.bash
5  admin.bash  check_dir.bash  define_tasks.bash  define_var.bash  exit_status1.bash  exit_status2.bash  error.txt  error_exist.bash  error_file.bash  hello_world.bash  list_of_users.bash  loop_output1.bash  loop_output2.bash  perform_task.bash  read_input.bash  response_output.bash  sample      sample_input.bash  sample_input2.bash  sequence.bash  specify_days.bash  variable.bash  variable_value_met.bash
[root@ip-172-31-25-75 scripts]# |
```

4. Create a bash script to **install Nginx on an EC2 server.**

- ➔ Created a bash file
- ➔ Used “yum install nginx -y” for nginx installation
- ➔ Enable >> systemctl enable nginx
- ➔ Start >> systemctl start nginx
- ➔ Status >> systemctl status nginx
- ➔ Checked the status and nginx is active and running

```
[root@ip-172-31-25-75 ~]# cd /scripts
[root@ip-172-31-25-75 scripts]# vi install_nginx.bash
[root@ip-172-31-25-75 scripts]# chmod 755 install_nginx.bash
[root@ip-172-31-25-75 scripts]# cat install_nginx.bash
#!/bin/bash

sudo yum install nginx -y
sudo systemctl enable nginx
sudo systemctl start nginx

echo "Nginx is installed and started successfully"

sudo systemctl status nginx --no-pager
[root@ip-172-31-25-75 scripts]# ./install_nginx.bash
last metadata expiration check: 0:06:38 ago on Mon Aug 17 02:52:10 2026.
Package nginx-1:1.30.4-1.amzn2023.0.1.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service → /usr/lib/systemd/system/nginx.service.
Nginx is installed and started successfully
● nginx.service - The nginx HTTP and reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: disabled)
   Active: active (running) since Mon 2026-08-17 02:58:49 UTC; 6ms ago
     Process: 2348 ExecStartPre=/usr/bin/rm -f /run/nginx.pid (code=exited, status=0/SUCCESS)
     Process: 2349 ExecStartPre=/usr/sbin/nginx -t (code=exited, status=0/SUCCESS)
     Process: 2350 ExecStart=/usr/sbin/nginx (code=exited, status=0/SUCCESS)
    Main PID: 2351 (nginx)
         Tasks: 3 (limit: 1059)
        Memory: 5.9M
           CPU: 60ms
      CGroup: /system.slice/nginx.service
              └─2351 "nginx: master process /usr/sbin/nginx"
                  └─2352 "nginx: worker process"
                      └─2353 "nginx: worker process"

Aug 17 02:58:49 ip-172-31-25-75.ec2.internal systemd[1]: Starting nginx.service - The nginx HTTP and reverse proxy server...
Aug 17 02:58:49 ip-172-31-25-75.ec2.internal nginx[2349]: nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
Aug 17 02:58:49 ip-172-31-25-75.ec2.internal nginx[2349]: nginx: configuration file /etc/nginx/nginx.conf test is successful
Aug 17 02:58:49 ip-172-31-25-75.ec2.internal systemd[1]: Started nginx.service - The nginx HTTP and reverse proxy server.
[root@ip-172-31-25-75 scripts]#
```

5. Create a bash script to **install Apache Tomcat on an EC2 server.**

- ➔ Created a bash file.
- ➔ Installed Java using the package manager.
- ➔ Downloaded the Apache Tomcat archive using “wget”
- ➔ Extracted Tomcat under /opt.
- ➔ Renamed the extracted directory to tomcat.
- ➔ Started Tomcat using startup.sh.

```
[root@ip-172-31-25-75 scripts]# ./tomcat_installation.bash
Installing Java...
last metadata expiration check: 3:56:36 ago on Mon Aug 17 02:52:10 2026.
Package java-17-amazon-corretto-1:17.0.20+8-1.amzn2023.1.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
Java installation successful.
Checking Java version...
openjdk version "17.0.20" 2026-07-21 LTS
OpenJDK Runtime Environment Corretto-17.0.20.8.1 (build 17.0.20+8-LTS)
OpenJDK 64-Bit Server VM Corretto-17.0.20.8.1 (build 17.0.20+8-LTS, mixed mode, sharing)
Java is working successfully.
Downloading Apache Tomcat...
--2026-08-17 06:48:46-- https://dlcdn.apache.org/tomcat/tomcat-9/v9.0.120/bin/apache-tomcat-9.0.120.tar.gz
Resolving dlcdn.apache.org (dlcdn.apache.org)... 151.101.2.132, 2a04:de42::644
Connecting to dlcdn.apache.org (dlcdn.apache.org)|151.101.2.132|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 13154363 (12M) [application/x-gzip]
Saving to: 'apache-tomcat-9.0.120.tar.gz.2'

apache-tomcat-9.0.120.tar.gz.2 100%[=====] 12.54M ---KB/s in 0.03s

2026-08-17 06:48:46 (444 MB/s) - 'apache-tomcat-9.0.120.tar.gz.2' saved [13154363/13154363]

Tomcat download successful.
Extracting Tomcat...
apache-tomcat-9.0.120/conf/
apache-tomcat-9.0.120/conf/catalina.policy
apache-tomcat-9.0.120/conf/catalina.properties
apache-tomcat-9.0.120/conf/context.xml
apache-tomcat-9.0.120/conf/jaspic-providers.xml
apache-tomcat-9.0.120/conf/jaspic-providers.xsd
apache-tomcat-9.0.120/conf/logging.properties
apache-tomcat-9.0.120/conf/server.xml
apache-tomcat-9.0.120/conf/tomcat-users.xml
apache-tomcat-9.0.120/conf/web.xml
apache-tomcat-9.0.120/bin/
apache-tomcat-9.0.120/lib/
apache-tomcat-9.0.120/logs/
apache-tomcat-9.0.120/temp/
apache-tomcat-9.0.120/webapps/
apache-tomcat-9.0.120/webapps/ROOT/
apache-tomcat-9.0.120/webapps/ROOT/WEB-INF/
apache-tomcat-9.0.120/webapps/docs/
apache-tomcat-9.0.120/webapps/docs/META-INF/
apache-tomcat-9.0.120/webapps/docs/WEB-INF/jsp/
apache-tomcat-9.0.120/webapps/docs/annotationapi/
apache-tomcat-9.0.120/webapps/docs/api/
apache-tomcat-9.0.120/webapps/docs/appdev/
apache-tomcat-9.0.120/webapps/docs/appdev/sample/
apache-tomcat-9.0.120/webapps/docs/appdev/sample/docs/
apache-tomcat-9.0.120/webapps/docs/appdev/sample/src/
apache-tomcat-9.0.120/webapps/docs/appdev/sample/src/mypackage/
apache-tomcat-9.0.120/webapps/docs/appdev/sample/web/
apache-tomcat-9.0.120/webapps/docs/appdev/sample/web/WEB-INF/
```

```
root@ip-172-31-25-75:~/jsoo/h/scripts
apache-tomcat-9.0.120/webapps/docs/appdev/sample/docs/
apache-tomcat-9.0.120/webapps/docs/appdev/sample/src/
apache-tomcat-9.0.120/webapps/docs/appdev/sample/src/mypackage/
apache-tomcat-9.0.120/webapps/docs/appdev/sample/web/
apache-tomcat-9.0.120/webapps/docs/appdev/sample/web/WEB-INF/
apache-tomcat-9.0.120/webapps/docs/appdev/sample/web/images/
apache-tomcat-9.0.120/webapps/docs/architecture/
apache-tomcat-9.0.120/webapps/docs/architecture/requestProcess/
apache-tomcat-9.0.120/webapps/docs/architecture/startup/
apache-tomcat-9.0.120/webapps/docs/config/
apache-tomcat-9.0.120/webapps/docs/elapi/
apache-tomcat-9.0.120/webapps/docs/images/
apache-tomcat-9.0.120/webapps/docs/images/fonts/
apache-tomcat-9.0.120/webapps/docs/jspicapi/
apache-tomcat-9.0.120/webapps/docs/jspapi/
apache-tomcat-9.0.120/webapps/docs/servletapi/
apache-tomcat-9.0.120/webapps/docs/tribes/
apache-tomcat-9.0.120/webapps/docs/websocketapi/
apache-tomcat-9.0.120/webapps/examples/META-INF/
apache-tomcat-9.0.120/webapps/examples/WEB-INF/
apache-tomcat-9.0.120/webapps/examples/WEB-INF/classes/
apache-tomcat-9.0.120/webapps/examples/WEB-INF/classes/async/
apache-tomcat-9.0.120/webapps/examples/WEB-INF/classes/checkbox/
apache-tomcat-9.0.120/webapps/examples/WEB-INF/classes/compressionfilters/
apache-tomcat-9.0.120/webapps/examples/WEB-INF/classes/dates/
apache-tomcat-9.0.120/webapps/examples/WEB-INF/classes/examples/
apache-tomcat-9.0.120/webapps/examples/WEB-INF/classes/filters/
apache-tomcat-9.0.120/webapps/examples/WEB-INF/classes/http2/
apache-tomcat-9.0.120/webapps/examples/WEB-INF/classes/jsp2/
apache-tomcat-9.0.120/webapps/examples/WEB-INF/classes/jsp2/examples/
apache-tomcat-9.0.120/webapps/examples/WEB-INF/classes/jsp2/examples/el/
apache-tomcat-9.0.120/webapps/examples/WEB-INF/classes/jsp2/examples/impltag/
apache-tomcat-9.0.120/webapps/examples/WEB-INF/classes/listeners/
apache-tomcat-9.0.120/webapps/examples/WEB-INF/classes/nonlocking/
apache-tomcat-9.0.120/webapps/examples/WEB-INF/classes/num/
apache-tomcat-9.0.120/webapps/examples/WEB-INF/classes/sessions/
apache-tomcat-9.0.120/webapps/examples/WEB-INF/classes/trackers/
apache-tomcat-9.0.120/webapps/examples/WEB-INF/classes/util/
apache-tomcat-9.0.120/webapps/examples/WEB-INF/classes/validators/
apache-tomcat-9.0.120/webapps/examples/WEB-INF/classes/websocket/
apache-tomcat-9.0.120/webapps/examples/WEB-INF/classes/websocket/chat/
apache-tomcat-9.0.120/webapps/examples/WEB-INF/classes/websocket/drawboard/
apache-tomcat-9.0.120/webapps/examples/WEB-INF/classes/websocket/drawboard/wsmessages/
apache-tomcat-9.0.120/webapps/examples/WEB-INF/classes/websocket/echo/
apache-tomcat-9.0.120/webapps/examples/WEB-INF/classes/websocket/snake/
apache-tomcat-9.0.120/webapps/examples/WEB-INF/jsp/
apache-tomcat-9.0.120/webapps/examples/WEB-INF/lib/
apache-tomcat-9.0.120/webapps/examples/WEB-INF/tags/
apache-tomcat-9.0.120/webapps/examples/jsp/
apache-tomcat-9.0.120/webapps/examples/jsp/async/
apache-tomcat-9.0.120/webapps/examples/jsp/checkbox/
apache-tomcat-9.0.120/webapps/examples/jsp/colors/
apache-tomcat-9.0.120/webapps/examples/jsp/dates/
apache-tomcat-9.0.120/webapps/examples/jsp/error/
apache-tomcat-9.0.120/webapps/examples/jsp/forward/
```

```
root@ip-172-31-25-75:~/jsoo/h/scripts
apache-tomcat-9.0.120/webapps/examples/websocket/chat.xhtml
apache-tomcat-9.0.120/webapps/examples/websocket/drawboard.xhtml
apache-tomcat-9.0.120/webapps/examples/websocket/echo.xhtml
apache-tomcat-9.0.120/webapps/examples/websocket/index.xhtml
apache-tomcat-9.0.120/webapps/examples/websocket/snake.xhtml
apache-tomcat-9.0.120/webapps/host-manager/META-INF/context.xml
apache-tomcat-9.0.120/webapps/host-manager/WEB-INF/jsp/401.jsp
apache-tomcat-9.0.120/webapps/host-manager/WEB-INF/jsp/403.jsp
apache-tomcat-9.0.120/webapps/host-manager/WEB-INF/jsp/404.jsp
apache-tomcat-9.0.120/webapps/host-manager/WEB-INF/manager.xml
apache-tomcat-9.0.120/webapps/host-manager/WEB-INF/web.xml
apache-tomcat-9.0.120/webapps/host-manager/css/manager.css
apache-tomcat-9.0.120/webapps/host-manager/images/ast-loop.svg
apache-tomcat-9.0.120/webapps/host-manager/images/favicon.ico
apache-tomcat-9.0.120/webapps/host-manager/images/tomcat.svg
apache-tomcat-9.0.120/webapps/host-manager/index.jsp
apache-tomcat-9.0.120/webapps/manager/META-INF/context.xml
apache-tomcat-9.0.120/webapps/manager/WEB-INF/jsp/401.jsp
apache-tomcat-9.0.120/webapps/manager/WEB-INF/jsp/403.jsp
apache-tomcat-9.0.120/webapps/manager/WEB-INF/jsp/404.jsp
apache-tomcat-9.0.120/webapps/manager/WEB-INF/jsp/connectorCerts.jsp
apache-tomcat-9.0.120/webapps/manager/WEB-INF/jsp/connectorTrustedCerts.jsp
apache-tomcat-9.0.120/webapps/manager/WEB-INF/jsp/sessionDetail.jsp
apache-tomcat-9.0.120/webapps/manager/WEB-INF/jsp/sessionsList.jsp
apache-tomcat-9.0.120/webapps/manager/WEB-INF/web.xml
apache-tomcat-9.0.120/webapps/manager/css/manager.css
apache-tomcat-9.0.120/webapps/manager/images/ast-loop.svg
apache-tomcat-9.0.120/webapps/manager/images/favicon.ico
apache-tomcat-9.0.120/webapps/manager/images/tomcat.svg
apache-tomcat-9.0.120/webapps/manager/index.jsp
apache-tomcat-9.0.120/webapps/manager/status.xsd
apache-tomcat-9.0.120/webapps/manager/xform.xml
apache-tomcat-9.0.120/bin/catalina.sh
apache-tomcat-9.0.120/bin/ciphers.sh
apache-tomcat-9.0.120/bin/configtest.sh
apache-tomcat-9.0.120/bin/daemon.sh
apache-tomcat-9.0.120/bin/digest.sh
apache-tomcat-9.0.120/bin/mkwebase.sh
apache-tomcat-9.0.120/bin/setclasspath.sh
apache-tomcat-9.0.120/bin/shutdown.sh
apache-tomcat-9.0.120/bin/startup.sh
apache-tomcat-9.0.120/bin/tool-wrapper.sh
apache-tomcat-9.0.120/bin/version.sh
Tomcat extraction successful.
Remaining Tomcat directory...
Setting permissions...
Starting Tomcat...
Using CATALINA_BASE: /opt/tomcat
Using CATALINA_HOME: /opt/tomcat
Using CATALINA_TMPDIR: /opt/tomcat/temp
Using JRE_HOME: /usr
Using CLASSPATH: /opt/tomcat/bin/bootstrap.jar:/opt/tomcat/bin/tomcat-juli.jar
Tomcat started.
Tomcat started successfully.
Tomcat installation completed successfully.
[root@ip-172-31-25-75: scripts]#
```

6. Create a bash script to **check if the Nginx service is running**, if not running then script should start the service.

- ➔ Created a bash file
- ➔ Checked the status using “systemctl is-active --quiet nginx”
- ➔ Used if else to check the status and update.
- ➔ Used commands:
 - systemctl start nginx
 - systemctl status nginx

```

root@ip-172-31-25-75:~/poojith/scripts
[root@ip-172-31-25-75 scripts]# cat check_nginx.bash
#!/bin/bash

if systemctl is-active -quiet nginx
then
    echo "Nginx is running"
else
    echo "Nginx is not running, Starting Nginx"
    sudo systemctl start nginx
    echo "nginx is started"
    sudo systemctl status nginx
    echo "Nginx is running"
fi
[root@ip-172-31-25-75 scripts]# ./check_nginx.bash
systemctl: invalid option -- 'u'
Nginx is not running, Starting Nginx
nginx is started
● nginx.service - The nginx HTTP and reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: disabled)
   Active: active (running) since Mon 2026-08-17 03:05:05 UTC; 41s ago
     Process: 2594 ExecStartPre=/usr/bin/rm -f /run/nginx.pid (code=exited, status=0/SUCCESS)
     Process: 2601 ExecStartPre=/usr/sbin/nginx -t (code=exited, status=0/SUCCESS)
     Process: 2617 ExecStart=/usr/sbin/nginx (code=exited, status=0/SUCCESS)
    Main PID: 2651 (nginx)
      Tasks: 3 (limit: 1059)
     Memory: 3.4M
        CPU: 77ms
    CGroup: /system.slice/nginx.service
            └─2651 "nginx: master process /usr/sbin/nginx"
              └─2652 "nginx: worker process"
                └─2653 "nginx: worker process"

Aug 17 03:05:04 ip-172-31-25-75.ec2.internal systemd[1]: Starting nginx.service - The nginx HTTP and reverse proxy server...
Aug 17 03:05:04 ip-172-31-25-75.ec2.internal nginx[2601]: nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
Aug 17 03:05:04 ip-172-31-25-75.ec2.internal nginx[2601]: nginx: configuration file /etc/nginx/nginx.conf test is successful
Aug 17 03:05:05 ip-172-31-25-75.ec2.internal systemd[1]: Started nginx.service - The nginx HTTP and reverse proxy server.
Nginx is running
[root@ip-172-31-25-75 scripts]#

```

7. Create a bash script for a **calculator**.

- ➔ Created a bash file
- ➔ Used read command to take the input from the user
- ➔ Used case operator to check multiple possible values
- ➔ Used arithmetic operators to perform calculations.
- ➔ Executed and verified the results.

```

root@ip-172-31-25-75:~/poojith/scripts
[root@ip-172-31-25-75 scripts]# vi Calculator.bash
[root@ip-172-31-25-75 scripts]# chmod 755 Calculator.bash
[root@ip-172-31-25-75 scripts]# cat Calculator.bash
#!/bin/bash

echo "Enter first number:"
read num1

echo "Enter second number:"
read num2

echo "Enter operation (+, -, *, /):"
read operator

case $operator in
    +)
        result=$((num1 + num2))
        ;;
    -)
        result=$((num1 - num2))
        ;;
    \*)
        result=$((num1 * num2))
        ;;
    /)
        result=$((num1 / num2))
        ;;
    *)
        echo "Invalid operator"
        exit 1
        ;;
esac

echo "Result: $result"
[root@ip-172-31-25-75 scripts]# ./Calculator.bash
Enter first number:
12
Enter second number:
15
Enter operation (+, -, *, /):
+
Result: 27
[root@ip-172-31-25-75 scripts]# ./Calculator.bash
Enter first number:
5
Enter second number:
8
Enter operation (+, -, *, /):
*
Result: 40
[root@ip-172-31-25-75 scripts]#

```

8. Create a bash script to **check if a directory exists**, if not then create a directory.

➔ Created a bash file

➔

```
root@ip-172-31-25-75:~/poojith/scripts
[root@ip-172-31-25-75 scripts]# cat make_DIR.bash
#!/bin/bash

DIR="/root/poojith/scripts/RAVUTLA"

if [ -d "$DIR" ]
then
    echo "Directory already exists."
else
    mkdir -p "$DIR"
    echo "Directory created successfully."
fi

[root@ip-172-31-25-75 scripts]# ./make_DIR.bash
Directory created successfully.
```

9. Create a bash script to **delete the last 3 lines of a file**.

➔ Created a bash file

➔ Used head -n -3 to remove the last three lines.

➔ Stored the modified content in a temporary file.

➔ Replaced the original file with the modified file.

➔ Executed and verified the output using cat.

```
[root@ip-172-31-25-75 scripts]# vi file1.txt
[root@ip-172-31-25-75 scripts]# vi rm_last3.bash
[root@ip-172-31-25-75 scripts]# chmod 755 rm_last3.bash
[root@ip-172-31-25-75 scripts]# cat rm_last3.bash
#!/bin/bash

file="file1.txt"

head -n -3 "$file" > temp.txt && mv temp.txt "$file"

echo "Last 3 lines deleted successfully."
[root@ip-172-31-25-75 scripts]# ./rm_last3.bash
Last 3 lines deleted successfully.
[root@ip-172-31-25-75 scripts]# cat file1.txt
line 1
line 2
line 3
line 4
[root@ip-172-31-25-75 scripts]#
```

10. Bash script to monitor cpu and if it is more than 80% then send email notification.

➔ Created a bash fil.

➔ checked CPU utilization.

➔ Extracted the usage percentage.

➔ Compared the usage with the 80% threshold.

➔ Sent an email notification when usage exceeded 80%.

➔ Modified permissions using chmod.

➔ Executed the script and verified cpu usage.

```
ec2-user@ip-172-31-25-75:~$ vi cpu_usage.bash
[ec2-user@ip-172-31-25-75 ~]$ ./cpu_usage.bash
[ec2-user@ip-172-31-25-75 ~]$ chmod 755 cpu_usage.bash
[ec2-user@ip-172-31-25-75 ~]$ cat cpu_usage.bash
#!/bin/bash
# Purpose: checking the CPU usage, and more than 80, it will send email to user
# Date: 17/08/2026
# Modification: 17/08/2026
# Owner: Poojith

# CPU usage threshold
THRESHOLD=80

# Email address
EMAIL="yourmail@gmail.com"

# Get hostname
HOSTNAME=$(hostname)

# Calculate CPU usage
CPU_USAGE=$(top -bn1 | grep "Cpu(s)" | awk '{print 100 - %8}')

# Remove decimal part
CPU_USAGE=${CPU_USAGE%.*}

echo "Current CPU usage: ${CPU_USAGE}%"

# Check CPU usage
if [ "$CPU_USAGE" -gt "$THRESHOLD" ]; then

    SUBJECT="HIGH CPU USAGE - $HOSTNAME"

    MESSAGE="WARNING!
Server: $HOSTNAME
CPU Usage: ${CPU_USAGE}%

CPU usage has reached or exceeded the ${THRESHOLD}% threshold.
Please check the server."

    echo "$MESSAGE" | mail -s "$SUBJECT" "$EMAIL"

    if [ $? -eq 0 ]; then
        echo "Email notification sent successfully."
    else
        echo "Failed to send email notification."
        exit 1
    fi
fi

else
    echo "CPU usage is below ${THRESHOLD}%. No email notification required."
fi
[ec2-user@ip-172-31-25-75 ~]$ ./cpu_usage.bash
Current CPU usage: 100%
[ec2-user@ip-172-31-25-75 ~]$ ./cpu_usage.bash
./cpu_usage.bash: line 37: mail: command not found
Failed to send email notification.
[ec2-user@ip-172-31-25-75 ~]$ sudo yum install mailx -y
Last metadata expiration check: 1:11:47 ago on Mon Aug 17 02:52:10 2026.
Dependencies resolved.
```

```
[ec2-user@ip-172-31-25-75 ~]$ ./cpu_usage.bash
Current CPU usage: 100%
./cpu_usage.bash: line 37: mail: command not found
Failed to send email notification.
[ec2-user@ip-172-31-25-75 ~]$ sudo yum install mailx -y
Last metadata expiration check: 1:11:47 ago on Mon Aug 17 02:52:10 2026.
Dependencies resolved.
=====
Package                Architecture      Version           Repository        Size
-----
Installing:
mailx                  x86_64            12.5-43.amzn2023.0.1  amazonlinux      263 k
Transaction Summary
-----
Install 1 Package
Total download size: 263 k
Installed size: 497 k
Downloading Packages:
mailx-12.5-43.amzn2023.0.1.x86_64.rpm                                5.7 MB/s | 263 kB  00:00
Total                                                                2.9 MB/s | 263 kB  00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing                :
  Running scriptlet: mailx-12.5-43.amzn2023.0.1.x86_64                1/1
  Installing              : mailx-12.5-43.amzn2023.0.1.x86_64                1/1
  Running scriptlet: mailx-12.5-43.amzn2023.0.1.x86_64                1/1
  Verifying                : mailx-12.5-43.amzn2023.0.1.x86_64                1/1
Installed:
  mailx-12.5-43.amzn2023.0.1.x86_64
Complete!
[ec2-user@ip-172-31-25-75 ~]$ which mail
/usr/bin/mail
[ec2-user@ip-172-31-25-75 ~]$ ./cpu_usage.bash
Current CPU usage: 3%
CPU usage is below 80%. No email notification required.
[ec2-user@ip-172-31-25-75 ~]$
```

11. Bash script to monitor disk space and if it is more than 80% then send email notification.

- ➔ Created a bash fil.
- ➔ checked disk usage.
- ➔ Extracted the usage percentage.
- ➔ Compared the usage with the 80% threshold.
- ➔ Sent an email notification when usage exceeded 80%.
- ➔ Modified permissions using chmod.
- ➔ Executed the script and verified disk usage.

```
ec2-user@ip-172-31-25-75:~  
[ec2-user@ip-172-31-25-75 ~]$ vi disk_usage.bash  
[ec2-user@ip-172-31-25-75 ~]$ chmod 755 disk_usage.bash  
[ec2-user@ip-172-31-25-75 ~]$ cat disk_usage.bash  
#!/bin/bash  
  
# Purpose: Checking for disk usage and more than 80 it will send email to user  
# Date: 17/08/2026  
# Owner: Poojith  
  
# Disk usage threshold  
THRESHOLD=80  
  
# Email address  
EMAIL="yourmail@gmail.com"  
  
# Get disk usage of root filesystem  
DISK_USAGE=$(df / | awk 'NR==2 {print $5}' | tr -d '%')  
  
# Get hostname  
HOSTNAME=$(hostname)  
  
echo "Current disk usage: ${DISK_USAGE}%"  
  
# Check whether disk usage is greater than threshold  
if [ "$DISK_USAGE" -gt "$THRESHOLD" ]  
then  
    SUBJECT="HIGH DISK USAGE ALERT - $HOSTNAME"  
  
    MESSAGE="WARNING!  
Server: $HOSTNAME  
Root filesystem disk usage: ${DISK_USAGE}%  
Disk usage has reached or exceeded the ${THRESHOLD}% threshold.  
Please check the server."  
  
    echo "$MESSAGE" | mail -s "$SUBJECT" "$EMAIL"  
  
    if [ $? -eq 0 ]  
    then  
        echo "Email notification sent successfully."  
    else  
        echo "Failed to send email notification."  
        exit 1  
    fi  
else  
    echo "Disk usage is normal: ${DISK_USAGE}%"  
fi  
[ec2-user@ip-172-31-25-75 ~]$ ./disk_usage.bash  
Current disk usage: 26%  
Disk usage is normal: 26%  
[ec2-user@ip-172-31-25-75 ~]$ |
```

12. Bash script to monitor memory and if it is more than 80% then send email notification.

- ➔ Created a bash fil.
- ➔ checked memory usage.
- ➔ Extracted the usage percentage.
- ➔ Compared the usage with the 80% threshold.
- ➔ Sent an email notification when usage exceeded 80%.
- ➔ Modified permissions using chmod.
- ➔ Executed the script and verified memory usage.

```

ec2-user@ip-172-31-25-75:~
ec2-user@ip-172-31-25-75 ~]$ vi memory_monitor.bash
ec2-user@ip-172-31-25-75 ~]$ chmod 755 memory_monitor.bash
ec2-user@ip-172-31-25-75 ~]$ cat memory_monitor.bash
#!/bin/bash

# Purpose: Checking for memory usage and more than 80 it will send email to user
# Date: 17/08/2026
# Owner: Poojith

# Memory usage threshold
THRESHOLD=80

# Email address
EMAIL="ravutla.10@gmail.com"

# Get memory usage
MEMORY_USAGE=$(free | awk '/Mem:/ {printf "%.0f", ($3/$2)*100}')

# Get hostname
HOSTNAME=$(hostname)

echo "Current memory usage: ${MEMORY_USAGE}%"

# Check whether memory usage is greater than threshold
if [ "$MEMORY_USAGE" -gt "$THRESHOLD" ]
then
    SUBJECT="HIGH MEMORY USAGE ALERT - $HOSTNAME"

    MESSAGE="WARNING!
server: $HOSTNAME
memory usage: ${MEMORY_USAGE}%
memory usage has reached or exceeded the ${THRESHOLD}% threshold.
Please check the server."

    echo "$MESSAGE" | mail -s "$SUBJECT" "$EMAIL"

    if [ $? -eq 0 ]
    then
        echo "Email notification sent successfully."
    else
        echo "Failed to send email notification."
        exit 1
    fi
else
    echo "Memory usage is normal: ${MEMORY_USAGE}%"
fi
ec2-user@ip-172-31-25-75 ~]$ ./memory_monitor.bash
Current memory usage: 28%
Memory usage is normal: 28%
ec2-user@ip-172-31-25-75 ~]$

```


Exercise: Crontab Entries

1. April 5th Midnight
2. 5th of Every November, January, June if it is a Thursday
3. At 05 and 27th minutes of 9, 10, 11 hours every day
4. 34 minutes of 9th hour on 15th August
5. Every midnight
6. Every Weekend (Saturday night 11:59 PM)
7. After every reboot

```
root@ip-172-31-25-75:~/poojith/scripts
1 A..F      \.
2 B         a
3 C         admin.bash
4 Calculator.bash b
5 D         backup.bash
6 E         basic_second.bash
A RAVUTLA   c
[root@ip-172-31-25-75 scripts]# vi cronjob_test.bash
[root@ip-172-31-25-75 scripts]# crontab -l
# April 5 midnight
0 0 5 4 * /root/poojith/scripts/cronjob_test.bash

# 5th of January, June, November only if it is Thursday
0 0 5 1,6,11 * [ "$(date +%u)" -eq 4 ] && /root/poojith/scripts/cronjob_test.bash

# At 05 and 27 minutes of 9, 10, 11 hours every day
5,27 9,10,11 * * * /root/poojith/scripts/cronjob_test.bash

# August 15 at 9:34 AM
34 9 15 8 * /root/poojith/scripts/cronjob_test.bash

# Every midnight
0 0 * * * /root/poojith/scripts/cronjob_test.bash

# Every Saturday at 11:59 PM
59 23 * * 6 /root/poojith/scripts/cronjob_test.bash

# After every reboot
0 reboot /root/poojith/scripts/cronjob_test.bash
[root@ip-172-31-25-75 scripts]# pwd
/root/poojith/scripts
[root@ip-172-31-25-75 scripts]#
```